Programmable High-Power Optical Waveform Synthesis for Material Processing Using a Phase-Controlled Array of Spectrally Spaced Continuous-Wave Fiber Lasers

I am considering an array of **81 single-mode, 100 W continuous-wave (CW) fiber lasers**, each operating at a precisely defined Bragg wavelength between **1040.222 nm and 1069.900 nm**. Each laser has an **electro-optically adjustable phase**.

By appropriately controlling the relative phases and coherently combining the fiber outputs, the resulting waveform forms a **pulse train with pulses separated by 3 mm in propagation distance**. Each pulse has a temporal **full width at half maximum (FWHM) of approximately 100 femtoseconds** and a **peak power of approximately 0.8 MW**.

The phases of the individual lasers are **adjusted sequentially by measuring the output pulse peak power** and optimizing the phase of each channel.

Such a single-mode laser system could be used for **high-precision material processing**, combining high average power with ultrashort pulses.